

Extra Problems

Please work on the problems below and then discuss the problems at the [Week 6 Extra Problems Discussion](#) [↗ forum](#).

Extra problems week 6:

1. How do you win at misere nim? Misere nim is played just like normal nim, except that the person to take the last coin ON THE TABLE loses.
2. A subtraction game $\{a,b,c,\dots,d\}$, with positive integers $a,b,c,\dots,d$ is played with a heap of coins. Players can remove $a,b,c,\dots$, or d coins from any one pile. Let $G(n)$ be the Grundy number of the subtraction game given by $\{a,b,c,\dots,d\}$ with n coins to start. Come up with interesting sets $\{a,b,c,\dots,d\}$ such that the Grundy number is periodic. Come up with examples where the Grundy number is not periodic, but is if n is bigger than some integer M .
3. Let .007 represent the game: Remove 3 coins from a pile of coins, and optionally split into two piles. Let $G(n)$ denote the Grundy number of a pile of n coins to start. Is this periodic? Ultimately periodic (periodic for $n > M$). What about the game .077 (same, remove 2 or 3 coins)? And .777 (same remove 1,2, or 3 coins)?
4. One can come up with 1001 variations on 3, for instance, .004, remove 3 coins, as long as what is left is split into two non-zero piles. Study this, then make up some new ones, and study. Periodic? Ultimately periodic?

Mark as completed

